

SAMIRA HAMMOUCHE

Computer Engineering Student — Full-Stack & AI/ML Developer

Boumerdes, Algeria | samirahammouche17@gmail.com | +213 669 335 922 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Computer Engineering student specializing in Software Engineering with hands-on experience building full-stack web applications and AI-powered systems. Skilled in React, Next.js, TypeScript, Flask, FastAPI, PostgreSQL, and REST APIs, with practical experience in Machine Learning, Deep Learning, NLP, RAG, and LLM-based agents. Built projects including analytics dashboards, AI assistants, computer vision systems, and satellite-image disaster detection, with experience integrating AI capabilities into applications and deploying full-stack web applications.

EDUCATION

University M'hamed Bougara of Boumerdes (UMBB)

Engineering Degree in Software Engineering

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3

Web Development: React, Next.js, Flask, FastAPI, REST APIs, Responsive UI

Databases: PostgreSQL, SQL, Relational Database Design

AI / Machine Learning: Machine Learning, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), LLM Application Development, AI Agents, Scikit-learn, Pandas, NumPy, Computer Vision

Software Engineering & Tools: OOP, UML, MVC, Git/GitHub

EXPERIENCE

Web Engineering Intern | Faran Digital Academy (Remote) *August 2026 – September 2026 (2 months)*

- Built responsive web applications using React, Next.js, TypeScript, and Tailwind CSS.
- Developed an interactive analytics dashboard with REST API integration, search, filtering, and data visualization.

Advanced Software Engineering Job Simulation | Walmart (via Forage) *July 2026*

- Completed practical, simulation-based tasks in advanced data structures and software architecture as part of a Walmart engineering job simulation.
- Applied relational database design principles and performed data munging/cleaning exercises to prepare datasets for analysis.

IT Department Observership | Sonatrach *July 2025*

- Observed the IT department's work with SAP S/4HANA, gaining exposure to how enterprise ERP systems are integrated and used across a large organization's business processes.

PROJECTS

Churn Prediction System

Technologies: Python, Pandas, Scikit-learn, Logistic Regression, SVM

- Built a customer churn prediction pipeline by preprocessing historical customer data and training classification models, achieving 85% prediction accuracy and identifying the top 3 factors driving customer retention.

AI First Aid Assistant

Technologies: Python, NLP, Transformers, Sentence Embeddings, RAG

- Developed a conversational AI assistant for first-aid guidance by implementing intent recognition and a retrieval-augmented generation (RAG) pipeline over sentence embeddings, returning relevant emergency steps within seconds.

Watch Catalogue Web Application

Technologies: Flask, Python, HTML, CSS, JavaScript, SQL

- Designed and deployed a full-stack watch catalog platform with RESTful APIs and a relational database, supporting full CRUD operations through a responsive web interface.

Image Background Enhancer

Technologies: Python, OpenCV, NumPy, SDXL, Image Processing

- Built an application for automated image background enhancement, applying image processing techniques and SDXL-based generation to improve output quality.

InnovateX Platform (Group Project, 7-person team)

Technologies: UML, Requirements Analysis, System Design

- Led requirements analysis and UML workflow modeling for an innovation-matchmaking platform, producing a complete system design document used for prototype development.

CERTIFICATIONS

- Hugging Face — Agents Fundamentals Course
- Kaggle — Machine Learning Basics Course